Assumptions:

(1) Transmission occurs strictly through Aedes mosquito bites with no direct human-to-human spread.

(2) Once infected, mosquitoes do not recover and remain infectious for life

(3) Total human and mosquito populations remain approximately stable over the seasonal outbreak

(4) Personal protective measures reduce daily biting exposure, bounded by an empirical community compliance ceiling κ.

1. Governing Differential Equations

$$\begin{cases} \frac{dS_h}{dt} = d_h N_h - ab\frac{I_v}{N_h}S_h - d_h S_h \\ \frac{dE_h}{dt} = ab\frac{I_v}{N_h}S_h - (\epsilon_h + d_h)E_h \\ \frac{dI_h}{dt} = \epsilon_h E_h - (\gamma_h + d_h)I_h \\ \frac{dR_h}{dt} = \gamma_h I_h - d_h R_h \\ \frac{dS_v}{dt} = d_v N_v - ac\frac{I_h}{N_h}S_v - d_v S_v \\ \frac{dE_v}{dt} = ac\frac{I_h}{N_h}S_v - (\epsilon_v + d_v)E_v \\ \frac{dI_v}{dt} = \epsilon_v E_v - d_v I_v \end{cases}$$

1.1 Effective Biting Rate Parameterization

The effective mosquito biting rate a is formulated as:

$$a = a_0(1 - \kappa\eta)$$

where:

- a_0: Baseline daily biting frequency (bites · day^{-1} · mosquito^{-1})
- $\kappa \in [0,1]$: Community adherence ceiling to personal protective measures (dimensionless)
- $\eta \in [0,1]$: Physical barrier or chemical repellent efficacy per contact (dimensionless)

2. Total Population Balance Checks

$$\frac{dN_h}{dt} = \frac{dS_h}{dt} + \frac{dE_h}{dt} + \frac{dI_h}{dt} + \frac{dR_h}{dt} = d_h N_h - d_h(S_h + E_h + I_h + R_h) = d_h N_h - d_h N_h = 0 \Longrightarrow N_h(t) \equiv \text{const}$$

$$\frac{dN_v}{dt} = \frac{dS_v}{dt} + \frac{dE_v}{dt} + \frac{dI_v}{dt} = d_v N_v - d_v(S_v + E_v + I_v) = d_v N_v - d_v N_v = 0 \Longrightarrow N_v(t) \equiv mN_h \equiv \text{const}$$

3. State Vector and Disease-Free Equilibrium (DFE)

$$X = (S_h, E_h, I_h, R_h, S_v, E_v, I_v)^T$$

$$E_h = 0, \quad I_h = 0, \quad R_h = 0, \quad E_v = 0, \quad I_v = 0$$
$$d_h N_h - d_h S_h = 0 \Longrightarrow S_h^* = N_h$$
$$d_v N_v - d_v S_v = 0 \Longrightarrow S_v^* = N_v = mN_h$$

$$\boldsymbol{\mathcal{E}_0 = (N_h, 0, 0, 0, mN_h, 0, 0)}$$

4. Next Generation Matrix Formulation

4.1 Infected Subsystem Ordering

$$x = \begin{pmatrix} E_h \\ I_h \\ E_v \\ I_v \end{pmatrix}$$

$$\dot{x} = \mathcal{F}(x) - \mathcal{V}(x)$$

4.2 Transmission and transitional vector

$$\mathcal{F}(x) = \begin{pmatrix} ab\dfrac{I_v}{N_h}S_h \\ 0 \\ ac\dfrac{I_h}{N_h}S_v \\ 0 \end{pmatrix}, \quad \mathcal{V}(x) = \begin{pmatrix} (\epsilon_h + d_h)E_h \\ -\epsilon_h E_h + (\gamma_h + d_h)I_h \\ (\epsilon_v + d_v)E_v \\ -\epsilon_v E_v + d_v I_v \end{pmatrix}$$

4.3 Jacobian Evaluation at DFE

$$F = \left[\frac{\partial \mathcal{F}_i}{\partial x_j}(\mathcal{E}_0)\right] = \begin{pmatrix} 0 & 0 & 0 & \dfrac{abS_h}{N_h}\Big|_{\mathcal{E}_0} \\ 0 & 0 & 0 & 0 \\ 0 & \dfrac{acS_v}{N_h}\Big|_{\mathcal{E}_0} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & ab \\ 0 & 0 & 0 & 0 \\ 0 & ac \cdot m & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$V = \left[\frac{\partial \mathcal{V}_i}{\partial x_j}(\mathcal{E}_0)\right] = \begin{pmatrix} \epsilon_h + d_h & 0 & 0 & 0 \\ -\epsilon_h & \gamma_h + d_h & 0 & 0 \\ 0 & 0 & \epsilon_v + d_v & 0 \\ 0 & 0 & -\epsilon_v & d_v \end{pmatrix}$$

5. Matrix Version of V

$$V = \begin{pmatrix} V_h & 0 \\ 0 & V_v \end{pmatrix}, \quad V_h = \begin{pmatrix} \epsilon_h + d_h & 0 \\ -\epsilon_h & \gamma_h + d_h \end{pmatrix}, \quad V_v = \begin{pmatrix} \epsilon_v + d_v & 0 \\ -\epsilon_v & d_v \end{pmatrix}$$

$$V_h^{-1} = \frac{1}{(\epsilon_h + d_h)(\gamma_h + d_h)} \begin{pmatrix} \gamma_h + d_h & 0 \\ \epsilon_h & \epsilon_h + d_h \end{pmatrix} = \begin{pmatrix} \frac{1}{\epsilon_h + d_h} & 0 \\ \frac{\epsilon_h}{(\epsilon_h + d_h)(\gamma_h + d_h)} & \frac{1}{\gamma_h + d_h} \end{pmatrix}$$

$$V^{-1} = \begin{pmatrix} \frac{1}{\epsilon_h + d_h} & 0 & 0 & 0 \\ \frac{\epsilon_h}{(\epsilon_h + d_h)(\gamma_h + d_h)} & \frac{1}{\gamma_h + d_h} & 0 & 0 \\ 0 & 0 & \frac{1}{\epsilon_v + d_v} & 0 \\ 0 & 0 & \frac{\epsilon_v}{(\epsilon_v + d_v)d_v} & \frac{1}{d_v} \end{pmatrix}$$

6. Computation of Next Generation $K = FV^{-1}$

$$K = \begin{pmatrix} 0 & 0 & 0 & ab \\ 0 & 0 & 0 & 0 \\ 0 & ac \cdot m & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\epsilon_h + d_h} & 0 & 0 & 0 \\ \frac{\epsilon_h}{(\epsilon_h + d_h)(\gamma_h + d_h)} & \frac{1}{\gamma_h + d_h} & 0 & 0 \\ 0 & 0 & \frac{1}{\epsilon_v + d_v} & 0 \\ 0 & 0 & \frac{\epsilon_v}{(\epsilon_v + d_v)d_v} & \frac{1}{d_v} \end{pmatrix}$$

$$K = \begin{pmatrix} 0 & 0 & \frac{ab\epsilon_v}{(\epsilon_v + d_v)d_v} & \frac{ab}{d_v} \\ 0 & 0 & 0 & 0 \\ \frac{ac \cdot m \cdot \epsilon_h}{(\epsilon_h + d_h)(\gamma_h + d_h)} & \frac{ac \cdot m}{\gamma_h + d_h} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

7. Characteristic Equation and Spectral Radius $\rho(K)$

$$\det(\lambda I - K) = \det \begin{pmatrix} \lambda & 0 & -K_{13} & -K_{14} \\ 0 & \lambda & 0 & 0 \\ -K_{31} & -K_{32} & \lambda & 0 \\ 0 & 0 & 0 & \lambda \end{pmatrix} = 0$$

$$\det(\lambda I - K) = \lambda \cdot \det\begin{pmatrix} \lambda & 0 & -K_{13} \\ 0 & \lambda & 0 \\ -K_{31} & -K_{32} & \lambda \end{pmatrix} = \lambda \cdot \lambda \cdot \det\begin{pmatrix} \lambda & -K_{13} \\ -K_{31} & \lambda \end{pmatrix}$$

$$= \lambda^2(\lambda^2 - K_{13}K_{31}) = 0$$

$$\lambda^2\left(\lambda^2 - \left[\frac{ab\epsilon_v}{(\epsilon_v + d_v)d_v}\right]\left[\frac{ac \cdot m \cdot \epsilon_h}{(\epsilon_h + d_h)(\gamma_h + d_h)}\right]\right) = 0$$

$$\lambda_{1,2} = 0, \quad \lambda_{3,4} = \pm\sqrt{\frac{a^2 \cdot b \cdot c \cdot m \cdot \epsilon_h\epsilon_v}{(d_h + \epsilon_h)(d_h + \gamma_h)(d_v + \epsilon_v)d_v}}$$

8. Closed form expression for $\mathcal{R}_0$

$$\mathcal{R}_0 = \rho(K) = \max_i |\lambda_i|$$

$$\mathcal{R}_0 = \sqrt{\frac{a^2 \cdot b \cdot c \cdot m \cdot \epsilon_h\epsilon_v}{(d_h + \epsilon_h)(d_h + \gamma_h)(d_v + \epsilon_v)d_v}}$$